Project Proposal

Background and Motivation

Whether the substance of thought is reducible to physical processes remains an open question. Nevertheless, evidence undisputedly suggests that the organized interplay between neurons in the central nervous system gives rise to human sensory and cognitive capacity [1]. The first recording of what is now known as an action potential—the voltage across an electrically excitable cell's membrane as a function of time—was published in 1868 by Julius Bernstein [2]. Derived from frog tissue using a differential rheotome, an apparatus he invented, Bernstein's measurement provided pivotal evidence for theories regarding neural dynamics propounded by physicists and physiologists preceding him [2]. The interplay between neurons can, in turn, be studied by measuring the action potentials of individual neurons within and across anatomically defined brain regions [3].

A desire to measure action potentials at scale motivated the development of high-channel-count recording probes. The Neuropixels 1.0 probe, introduced in 2017, is an intracranial probe for extracellular electrophysiology recordings in rodents, sampling 384 channels simultaneously [3]. Its successor, the Neuropixels 2.0, extends this to as many as 5120 recording sites [4]. Such systems have enabled groundbreaking discoveries in systems neuroscience [5]. Higher channel counts, however, impose a proportionally higher bandwidth requirement.

The industry-standard analog front end for sampling raw extracellular voltage uses a 16-bit analog-to-digital converter (ADC) typically configured with a 30 kHz sampling rate [6, 7]. At these settings, a full Neuropixels 2.0 quad-base recording produces a data rate of 2.56 Gbps (see appendix for calculation). Bluetooth Low Energy 6.0 (BLE) is the state-of-the-art consumer wireless communication protocol with a power budget suitable for an implantable device [8,9]. In contrast to the data rate generated from a Neuropixels 2.0 recording, BLE has a theoretical maximum data rate of 4 Mbps, just 0.16 percent of the raw data rate generated [8]. Fortunately, many questions in neuroscientific inquiry require only action potential signatures, which constitute a small fraction of the continuous raw signal [10], [11].

Problem Statement

The bandwidth of Bluetooth Low Energy 6.0 is too low to support transmission of raw neural extracellular electrophysiology data sampled at the industry standard from more than two recording channels at once for offline spike sorting and consequent neuroscientific analysis [12].

Need Derived from Problem Statement

Compress continuous neural data on the recording head stage of a rodent for wireless transmission without fidelity loss of spatiotemporal spike information.

Design Requirements and Technical Specifications

Description

This section decomposes the stated need into the behaviors any valid solution must exhibit (the functional requirements) and the measurable qualities those behaviors must satisfy (the technical requirements). Both are stated independently of any particular design; the design approaches that could realize them are considered separately in the Design section.

Compressing continuous neural data without domain-specific loss of fidelity requires isolating and preserving exactly the domain-relevant events. the spikes, their identities, and their times—while discarding the remainder of the signal. Any solution must therefore perform five operations: it must *acquire* extracellular voltage from multiple recording channels, *detect* spike events within the acquired signal, *record* the time at which each detected spike occurs, *sort* each detected spike by assigning it a unit identity, and *transmit* each unit identity, paired with its timestamp, over a wireless link. These five operations constitute the functional requirements of the system. Each imposes one or more technical requirements—the measurable qualities its implementation must satisfy—grouped by the operation that produces them in Table 1.

These technical requirements were first derived independently for each operation. One requirement, however, resisted this per-operation decomposition: that no spike be lost. A sample or detected spike dropped anywhere in the chain—whether during acquisition, transmission to the sorter, or over the wireless link—compromises the final result identically. It is therefore treated as a system-level requirement spanning the full pipeline, summarized separately in Table 2, rather than attached to any single functional requirement in Table 1.

Consideration of Multiple Design Approaches

The functional and technical requirements above admit more than one design. The approaches considered here differ principally in *how* and *when* spikes are sorted—the operation that assigns each detected spike a unit identity. This single choice determines whether the system requires a separate offline phase at all.

Approach A: Offline-Supervised Sorting (Two-Phase)

This approach operates in two phases under different constraints. An offline phase, run once on a workstation, sorts a reference recording via unsupervised clustering and trains a classifier on the resulting labeled units. An online phase, running on the SoC, performs supervised inference against this pre-trained model. The offline phase is not subject to the power, latency, or memory constraints of the device; the online phase is. Its requirements therefore include the offline preparation as a precondition: a labeled unit library and trained classifier must exist before the device operates. [Systematic statement of the offline-phase operations and their requirements belongs here.]

Table 1: Per-operation technical requirements of the system.

Label	Requirement	Specification	Quantitative value	Measurement method
T1	≥ 3 channels amplified and digitized	channel count	≥ 3; target 32	count channels in acquired stream
T2	Sample fast enough for the spike band	per-channel sample rate	≥ 12.5 kS/s; design 25 kS/s	FFT of a known replayed signal
T3	Digitization keeps up across channels	aggregate conversion rate	$\geq$ channels $\times$ per-channel rate	sustained conversion count
T4	Resolve spikes above the noise floor	resolution / SNR	16-bit; SNR [TBD]	noise and SNR on a known input
T5	Link carries the digitized stream	ADC$\rightarrow$processor throughput	$\geq$ channels $\times$ rate $\times$ bits	dropped-sample count
T6	Spikes separated from the LFP	high-pass corner	$\sim$300 Hz	filter frequency response
T7	Detect spikes reliably	sensitivity / false-positive rate	TPR [TBD]; FPR [TBD]	compare to Kilosort 4.0
T8	A timestamp is recorded per spike	timestamp presence	one per detected spike	inspect output stream
T9	Each spike assigned a unit identity	unit identity	one of K units	inspect output
T10	Sort better than chance	accuracy vs. Kilosort	$> 1/K$ (chance); [TBD]	confusion matrix vs. Kilosort
T11	Packet carries identity and timestamp	packet content	5 B (1 B ID + 4 B ts)	inspect received packets

Approach B: Online-Unsupervised Sorting (Single-Phase)

This approach performs all sorting on the device as data arrive, using an online clustering algorithm (e.g., Osort). No offline phase and no reference recording are required; the system is self-contained and needs no per-subject calibration. Because clustering must proceed without the full data set available in advance, sorted-unit stability and accuracy are expected to be lower than the supervised approach—the central trade-off between the two designs.